

Control work 5, and work on the mistakes

The exponential block in one paper — and the drill that fixes the base-under-one trap for good.

LESSON 55–56

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 5

P2 · CHAPTER 2 REVIEW

LESSON CLOCK

3

40'

10'

25'

12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite	25 min
The base drill	12 min

LEARNING OBJECTIVES

- Apply the exponential methods under time.
- Decide the direction of an inequality from the base without prompting.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 221–224
Cambridge	Pure Mathematics 2 & 3, pp. 45–46
Workbook	Control paper E

01

Explanation

The paper — 25 marks, 40 minutes

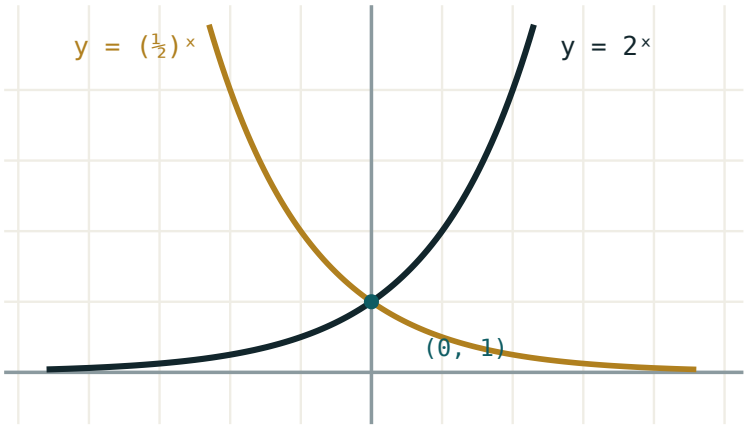
Q	TASK	MARKS	FROM
1	Sketch $y = 2^x - 4$, with asymptote, range and both intercepts	4	L49–50
2	Solve $3^{2x+1} = 27$ and $8^x = 4$	4	L51–52
3	Solve $4^x - 6 \cdot 2^x + 8 = 0$	4	L51–52
4	Solve $2^{x-1} > 32$ and $(\frac{1}{5})^x \geq 25$	5	L53–54
5	Solve $9^x - 10 \cdot 3^x + 9 < 0$	4	L53–54
6	A sample of 80 g has half-life 4 days. Find the mass after 12 days, and when 5 g remain	4	L49–50

Q4 CARRIES THE WHOLE TOPIC IN FIVE MARKS

Two for the first part, three for the second — of which one is for stating that the base is less than 1 and the sign must reverse.

The three errors

ERROR	LOOKS LIKE	KIND
sign not reversed	$(\frac{1}{2})^x > 8 \Rightarrow x > -3$	knowledge
$t > 0$ omitted	a negative t kept and turned into a false root	method
bases not matched	$8^x = 4$ attacked without writing $2^{3x} = 2^2$	method
power miscopied	$27 = 3^3$ written as 3^4	careless



both pass through $(0, 1)$; the x -axis is an asymptote

The picture that settles every direction question in this paper.

The base drill

Twelve inequalities on the board. For each, the class calls out only “**same**” or “**reverse**”, in two seconds:

INEQUALITY	CALL
$3^x > 9$	same
$(\frac{1}{3})^x > 9$	reverse
$0.7^x < 0.49$	reverse
$1.5^x \geq 2.25$	same
$(0.2)^x \leq 5$	reverse

ROW 5 IS THE ONE THEY MISS

$5 = (0.2)^{-1}$. Writing the right-hand side as a power of the **same** base is what makes the comparison possible at all.

02 Worked examples

EXAMPLE 1 Model answer, Q4b: solve $(\frac{1}{5})^x \geq 25$.

- ① Base $\frac{1}{5} < 1$ — the sign will reverse.

One mark for this line.

② $25 = (\frac{1}{5})^{-2}$

③ $x \leq -2$

ANSWER

$$x \leq -2$$

EXAMPLE 2 Model answer, Q5: solve $9^x - 10 \cdot 3^x + 9 < 0$.

- ① Let $t = 3^x > 0$.

$$t^2 - 10t + 9 < 0$$

② $(t - 1)(t - 9) < 0 \Rightarrow 1 < t < 9$

③ $3^0 < 3^x < 3^2$

Base > 1 , order preserved.

④ $0 < x < 2$

ANSWER

$$0 < x < 2$$

EXAMPLE 3 A learner wrote $(\frac{1}{2})^x > 8 \Rightarrow x > -3$. Name the mistake.

- ① The base is under 1.

- ② The function decreases, so the inequality reverses.
A knowledge gap.

- ③ Correct: $x < -3$.

ANSWER

Knowledge gap — the reversal rule

03 Interactive model

Run the base drill before the rewrite, and again at the end of the lesson.

Same or reverse?

$3^x > 9:$

$x > 2$

$x < 2$

$x > 9$

$x < 9$

$(\frac{1}{3})^x > 9:$

$x > 2$

$x < -2$

$x > -2$

$x < 2$

$8^x = 4:$

$x = 2$

$x = \frac{2}{3}$

$x = 0.5$

$x = \frac{3}{2}$

$$4^x - 6 \cdot 2^x + 8 = 0:$$

$$x = 1, 2$$

$$x = 2, 4$$

$$x = 0, 3$$

$$x = 2$$

$$2^{(x-1)} > 32:$$

$$x > 5$$

$$x > 6$$

$$x < 6$$

$$x > 4$$

80 g, half-life 4 days, after 12 days:

$$20 \text{ g}$$

$$10 \text{ g}$$

$$40 \text{ g}$$

$$5 \text{ g}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Common base	Umumiy asos	Общее основание
Reversing the sign	Ishorani almashtirish	Смена знака
Substitution	Almashtirish	Замена
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The direction of an exponential inequality is decided by:

the index

the base

the constant

the sign of x

A substitution must always carry:

$t \geq 0$

$t > 0$

$t \neq 0$

nothing

$8^x = 4$ is best done by:

guessing

writing both as powers of 2

dividing

squaring

Three half-lives leave:

$$\frac{1}{2}$$

$$\frac{1}{4}$$

$$\frac{1}{8}$$

$$\frac{1}{3}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $3^{(2x+1)} = 27$

ANSWER $x = 1$

2. Solve $8^x = 4$

ANSWER $x = \frac{2}{3}$

3. Solve $2^{(x-1)} > 32$

ANSWER $x > 6$

4. Solve $(\frac{1}{5})^x \geq 25$

ANSWER $x \leq -2$

5. Asymptote of $y = 2^x - 4$

ANSWER $y = -4$

6. x -intercept of $y = 2^x - 4$

ANSWER $x = 2$

7. 80 g, half-life 4 days, after 12 days

ANSWER 10 g

Medium · the standard question

7 PROBLEMS

1. Solve $4^x - 6 \cdot 2^x + 8 = 0$

ANSWER $x = 1, 2$

2. Solve $9^x - 10 \cdot 3^x + 9 < 0$

ANSWER $0 < x < 2$

3. Range of $y = 2^x - 4$

ANSWER $y > -4$

4. When do 5 g remain of 80 g with half-life 4 days?

ANSWER 16 days

5. Solve $25^x = 5^{(x+3)}$

ANSWER $x = 3$

6. Solve $(0.2)^x \leq 5$

ANSWER $x \geq -1$

7. Solve $2^{(x^2-1)} = 8$

ANSWER $x = \pm 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4^x - 2^{x+3} + 12 = 0$

ANSWER $x = 1$ and $x = \log_2 6$

2. Solve $9^x - 3^{x+2} + 8 \geq 0$

ANSWER $x \leq 0$ or $x \geq \log_3 8$

3. Solve $2^{2x+1} - 5 \cdot 2^x + 2 = 0$

ANSWER $x = -1, 1$

4. Sketch $y = 3^{x-2} + 1$: asymptote and y-intercept

ANSWER $y = 1; \frac{10}{9}$

5. A population trebles every 5 years. From 2000, when does it pass 50 000?

ANSWER ≈ 14.6 years

6. Solve $(\sqrt{2})^x = 8$

ANSWER $x = 6$

7. Solve $3^{2x} - 12 \cdot 3^x + 27 < 0$

ANSWER $1 < x < \log_3 9$ — i.e. $1 < x < 2$

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. State the base rule on every inequality.

1. Rewrite in full every question that lost a mark, with the base rule stated.
2. Five problems from the section your knowledge column was heaviest in.
3. Solve $16^x - 10 \cdot 4^x + 16 = 0$ and $(\frac{1}{7})^{x+1} > 49$.

4. A sample of 120 g has half-life 3 hours . Find the mass after 15 hours and the time until 7.5 g remain.